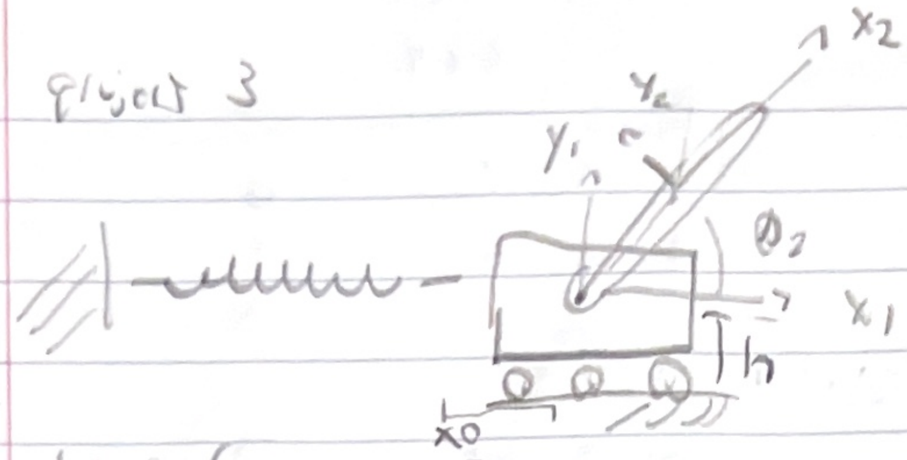
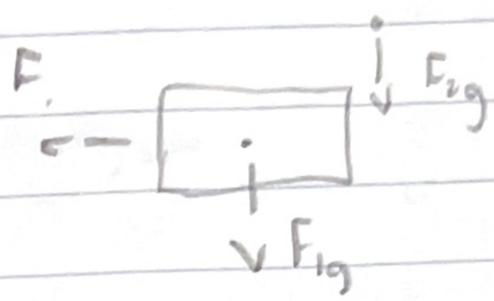
Project 3



External forces



block force $F = -k_x x_1$

gravitational force $F_{g1} = -m_1 g \hat{j}$ $F_{g2} = -m_2 g \hat{j}$

$m = 0.1$ kg

$m_2 = 0.3$ kg

Mass matrix =
$$\begin{bmatrix} m_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & m_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{m_1 l^2}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & m_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & m_2 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{12} m_2 (a^2 + b^2) \end{bmatrix}$$

a, b equal blocks dimensions

height and width

Constraint eqns:

① $y_1 - h - \frac{l}{2}\sin(\theta_1) = 0$ ②

③ $x_1 + \frac{l}{2}\cos(\theta_2) = x_2 \Rightarrow x_1 - x_2 + \frac{l}{2}\cos(\theta_1) = 0$

④ $y_1 + \frac{l}{2}\sin(\theta_1) = y_2 \Rightarrow y_1 - y_2 + \frac{l}{2}\sin(\theta_2) = 0$

Constraint matrix

$$\bar{C}_q \Rightarrow \begin{array}{c} \\ ① \\ ② \\ ③ \\ ④ \end{array} \begin{array}{c} \begin{array}{cccccc} x_1 & y_1 & \theta_1 & x_2 & y_2 & \theta_2 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 & -\frac{l}{2}\sin(\theta_2) \\ 0 & 1 & 0 & 0 & -1 & \frac{l}{2}\cos(\theta_2) \end{bmatrix} \end{array}$$

$$\bar{Q}_e = \begin{bmatrix} -k_x(x_1 - x_0) \\ -mg \\ 0 \\ 0 \\ -m_2 g \\ 0 \end{bmatrix}$$

$$\bar{Q}_i = -\bar{C}_t - (C_q \dot{q})_q \dot{q} - 2C_{qt}\dot{q} \quad \text{eqn } 1,2 = 0$$

$$C_t = \frac{d}{dt}(C(q,t)) = 0$$

$$(C_q \dot{q})_q \dot{q} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{l}{2}\cos\theta\dot{\theta} \\ 0 & 0 & 0 & 0 & 0 & -\frac{l}{2}\sin\theta\dot{\theta} \end{bmatrix} \begin{bmatrix} \dot{x}_1 \\ \dot{y}_1 \\ \dot{\theta}_1 \\ \dot{x}_2 \\ \dot{y}_1 \\ \dot{\theta}_2 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 \\ 0 \\ +\dot{\theta}_2^2\cos(\theta_2)\frac{l}{2} \\ +\dot{\theta}_2\sin(\theta_2)\frac{l}{2} \end{bmatrix}$$

Total $A\bar{x} = b$

$$
\begin{array}{c}
M \\ c\times 6 \\ \\ \bar{C}_q \\ 4\times 6
\end{array}
\left[\begin{array}{cccccc}
m_1 & 0 & 0 & 0 & 0 & 0 \\
0 & m_1 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{m_1 l^2}{12} & 0 & 0 & 0 \\
0 & 0 & 0 & m_2 & 0 & 0 \\
0 & 0 & 0 & 0 & m_2 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{m}{12}(a^2+b^2) \\
\hline
0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 \\
1 & 0 & 0 & -1 & 0 & -\frac{l}{2}\sin(\Phi_2) \\
0 & 1 & 0 & 0 & -1 & -\frac{l}{2}\cos(\Phi_2)
\end{array}\right]
\begin{array}{c}
\bar{C}_q^T \\ 6\times 4 \\
\left[\begin{array}{cccc}
0 & 0 & 1 & 0 \\
1 & 0 & 0 & 1 \\
0 & 1 & 0 & 0 \\
0 & 0 & -1 & 0 \\
0 & 0 & 0 & -1 \\
0 & 0 & -\frac{l}{2}\sin\Phi_2 & -\frac{l}{2}\cos(\Phi_2)
\end{array}\right] \\
\bar{0} \\
4\times 4
\end{array}
\quad \Sigma \rangle
$$

$$
= \left[\begin{array}{c}
\ddot{x}_1 \\ \ddot{y}_1 \\ \ddot{\Phi}_1 \\ \ddot{x}_2 \\ \ddot{y}_2 \\ \ddot{\Phi}_2 \\ \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4
\end{array}\right]
\quad
\left[\begin{array}{c}
-k(x_1 - l_0) \\ -m_1 g \\ 0 \\ 0 \\ -m_2 g \\ 0 \\ 0 \\ 0 \\ \frac{l}{2}\dot{\Phi}_2^2\cos(\Phi_2) \\ \frac{l}{2}\dot{\Phi}_2^2\sin(\Phi_2)
\end{array}\right]
$$

$\bar{x}$ 10×1 $\qquad$ b 10×1